

SUMMARY

Data Scientist with **3+ years** of experience developing **AI-driven solutions** to solve complex business challenges. Skilled in **Python, R, SQL, and machine learning techniques** such as **supervised, unsupervised learning, deep learning, and NLP**. Expertise in integrating AI agents and advanced technologies like **LLM, RAG, and VR** for stock market prediction and education platforms. Proficient in big data tools such as **Hadoop, Spark, and Hive**, and cloud platforms like **AWS, Azure, and GCP**. Adept at **statistical analysis, predictive modeling, and data visualization** with **Tableau and Power BI**, delivering actionable insights and leading cross-functional teams in AI and data-driven projects.

TECHNICAL SKILLS

Programming Languages: Python (NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, PyTorch, Hugging Face models, OpenAI API), R (ggplot2, dplyr, caret), **SQL** (MySQL, PostgreSQL, SQL Server), **Scala/Java** (for big data technologies), **JavaScript/TypeScript** (for front-end frameworks like React)

Data Manipulation & Analysis: Data Wrangling & Cleaning, Feature Engineering, Statistical Analysis, Exploratory Data Analysis (EDA), Sustainable AI practices, Feature Selection, Dimensionality Reduction

Machine Learning & AI: Supervised & Unsupervised Learning, Deep Learning (Neural Networks, CNNs, RNNs, GANs), Natural Language Processing (NLP), Time Series Analysis, Transfer Learning, Model Evaluation & Tuning (Cross-validation, Hyperparameter Tuning), **Retrieval-Augmented Generation (RAG)**, **Large Language Models (LLMs)**, **Langchain Agents**, Reinforcement Learning, Model Explainability (e.g., SHAP), **OpenAI API** integration

Big Data Technologies: Hadoop, Spark, Hive, HBase, Kafka, Flink, Storm, Elasticsearch

Data Visualization: Tableau, Power BI, Seaborn, Plotly, Matplotlib, D3.js (for interactive web visualizations), Dash (for dashboard development)

Cloud Platforms & AI Services : **AWS** (S3, Redshift, SageMaker, Lambda, EC2, Athena), **Azure** (Azure Data Lake, Azure Machine Learning, Azure Cognitive Services), **Google Cloud Platform (GCP)** (BigQuery, AI Platform, AutoML), **Hugging Face** Transformers, AI-powered APIs like Google AI and Microsoft AI

DevOps & MLOps: Docker, Kubernetes, Jenkins (CI/CD), Apache Airflow (for scheduling), MLflow (for model management), DVC (Data Version Control), **FastAPI** (for building APIs and services), **Langservers** (for serving LLMs in real-time environments)

Databases & Data Warehousing: **SQL** (MySQL, PostgreSQL, SQL Server), **NoSQL** (MongoDB, Cassandra, Neo4j), Data Warehousing (Snowflake, Redshift, BigQuery), In-memory Databases (Redis)

AI & Deep Learning Frameworks: TensorFlow, PyTorch, Keras, Hugging Face, LangChain (for orchestrating AI agents), ONNX (for model interoperability), FastAI, OpenAI Gym (for reinforcement learning)

Tools & Frameworks: Jupyter Notebooks, Git/GitHub (version control), Docker/Kubernetes (for deployment), PyCaret, Streamlit (for rapid app development), Hugging Face, TensorFlow Serving, Gradio, FastAPI, **React** (for interactive front-end development)

Business & Domain Knowledge A/B Testing & Experimentation, Business Intelligence (BI) tools, Data-Driven Decision Making, Predictive Analytics, Industry-specific knowledge (Finance, Healthcare, E-commerce, Retail), Ethical AI & Sustainable AI

Soft Skills: Strong Communication & Presentation Skills, Critical Thinking & Problem Solving, Stakeholder Collaboration, Team Leadership, Time Management & Attention to Detail, Mentoring in AI and Data Science Practices.

PROFESSIONAL EXPERIENCE

Pennsylvania State University | Malvern, Pennsylvania

Oct 2023 - Present

Research Assistant in the Engineering Division(Data Scientist)

- Led Stock Market Prediction Project Using LLMs:-** Spearheaded a project that integrated **historical stock data** with real-time **news sentiment analysis** using **Large Language Models (LLMs)**, improving stock price prediction accuracy by **15%**. Deployed the model on **CauldOps AI** for **scalable AI** operations and used **Langchain** for efficient workflow management.
- Enhanced Prediction Model with RAG and Real-Time Insights:-** Developed a stock prediction model combining RAG for real-time news, **LLM-generated insights**, and **numerical data**. Achieved an **R² score** of **0.85** and reduced the Mean Absolute Percentage Error (MAPE) by 12%, currently preparing research for publication of this idea.
- Innovative VR-Based Educational Platform:-** Created an educational platform that converts **speech-to-text** and **text-to-image** with **real-time feedback**, leveraging **VR technology**. Integrated **APIs** with **Langserver**, **FastAPI**, and Unity, utilizing **Whisper** for **speech recognition** and **Stability Model** for **VR image generation**.
- AI-Powered Learning with Real-Time Translation:-** Developed an immersive **AI-driven platform** for **real-time language translation** and **AI-generated content**, transforming **education environments** with **speech-to-text** and **visual feedback using VR**. Collaborating on publishing research that **highlights the transformative impact of AI and VR in education**.
- Generative AI Similarity Matrix for Stock Data:-** Led the development of a **foundational similarity matrix** using **Generative AI** to **analyze historical stock data** and **market trends**. This project deepened insights into stock market behavior, showcasing technological innovation with advanced machine learning frameworks.
- AI-Driven Financial Sentiment Analysis:-** Integrated LLMs and RAG for robust news sentiment analysis, boosting stock prediction accuracy by 8%. This advanced AI model showcased improvements in predictive power, with findings being prepared for publication to demonstrate AI's advantages in financial prediction.

Interactive Manpower Solution Pvt. Ltd (Decimal)

Ahmedabad, Gujarat

Data Analyst

Oct 2022 – July 2023

- Transformed raw data into impactful narratives by seamlessly weaving together **R, SQL, and Python**, crafting innovative models and visualizations that empower **stakeholders** to make informed, data-driven decisions.
- Applied advanced **statistical analysis** techniques to uncover trends and patterns in complex data, transforming raw numbers into meaningful insights that guide strategic **business decisions** and drive measurable outcomes.

- Leveraged cutting-edge **machine learning** and **AI** techniques to develop predictive models and intelligent algorithms, enabling data-driven solutions that optimize performance and drive innovation across diverse business domains.
- Leveraged **big data technologies**, including **Hadoop** and **Spark**, to process and analyze terabytes of data, resulting in a **40%** improvement in data processing speed and enabling real-time analytics for critical business decisions.
- Utilized **Tableau** to create dynamic visualizations and interactive dashboards that transform complex data sets into clear, actionable insights, enhancing stakeholder engagement and driving informed decision-making.
- Expertly utilized **AWS services**, including **S3** for scalable storage, Redshift for data warehousing, SageMaker for building and deploying machine learning models, and **Lambda** for serverless computing, to architect robust data solutions that enhance efficiency and drive analytics-driven decision-making.
- Leveraged strong **communication** and **presentation skills** to convey complex data insights to non-technical stakeholders, facilitating informed decision-making and fostering cross-functional collaboration.
- Applied **critical thinking** and **problem-solving** skills to identify data anomalies, resulting in a **20%** reduction in reporting errors and enabling more accurate business intelligence insights.

Fingertips Data Intelligence Solutions Pvt. Ltd
Data Science Intern

Ahmedabad, Gujarat
June 2021 – June 2022

- Expertly harnessed the power of **MySQL** and **Python** to design efficient data pipelines and conduct in-depth analyses, turning complex datasets into actionable insights that drive strategic decision-making.
- Enhanced **data integrity** by implementing robust data wrangling and cleaning techniques, improving the accuracy of predictive models and boosting overall analysis efficiency by **30%**.
- Developed advanced **deep learning** models for natural language processing, achieving a **25%** increase in accuracy for sentiment analysis tasks and enabling more effective insights from unstructured text data.
- Created interactive **Power BI** dashboards that visualized complex datasets, enhancing stakeholder insights and decision-making efficiency by **35%** through real-time data analysis and reporting.
- Implemented **Azure Data Lake** for scalable data storage and utilized **Azure Machine Learning** to build predictive models, resulting in a **30%** increase in forecasting accuracy and streamlined data processing workflows.
- Designed and implemented efficient data warehousing solutions using **Snowflake**, **Redshift**, and **BigQuery**, optimizing data storage and retrieval processes to support real-time analytics and empower data-driven decision-making across the organization.

EDUCATION

Pennsylvania State University, PA

Master of Science in Computer Science (Data Analytics)

LD College of Engineering, Ahmedabad, Gujarat

Bachelor of Engineering in Computer Engineering

PROJECT

- **Facial Emotion Recognition Using CNNs:** Developed a deep learning model using CNN to classify emotions from 34,000 facial images, achieving 75% training and 64% testing accuracy. Leveraged Python and NumPy for image processing and ReLU-activated layers to enhance detection accuracy, with the entire project documented on [GitHub](#).
- **Google Stock Price Prediction with RNN-LSTM:** Built an RNN LSTM model for predicting Google stock prices, achieving an R^2 score of 0.76. Prepared and standardized historical stock data using Numpy and Pandas, with detailed project documentation available on [GitHub](#).
- **Fraud Detection System Using LightGBM & XGBoost:** Engineered a fraud detection pipeline using LightGBM and XGBoost, achieving high accuracy with strong F1 and AUC scores. Extensive feature engineering and data analysis were conducted on financial transaction data, and the full implementation is shared on [GitHub](#).

EXTRACURRICULAR ACTIVITIES

- Advanced to the second round of the Humana Competition in September 2023, ranking in the top 50 out of 200 teams, and collaborated in a trio for the [case study](#).
- We developed a multilingual chatbot for farmers, capable of answering questions and providing forecast predictions. The project won the local event and advanced to the global finalist stage. [Git](#).